THESIS PROJECT
TITLE PROPOSAL
(CS0025 – Software Engineering 1)

Group Name: **Statisfied**	Program: **Bachelor of Science in Computer Science with specialization in Data Science**	
Member's Name: 1. Dawn Andrei Pamesa (Project Manager) 2. Rogelio Porras (Member 1) 3. John Ashley Alba (Member 2) 4. Ralph Blanco (Member 3)	Term: [✓] 1st [ ] 2nd [ ] 3rd	Academic Year: 2026 - 2027
Proposed Project Title 2: **Project Cost Estimation and Overrun Prediction for a Quezon City Construction Firm Using Gradient Boosting**		

1.0 Area of Investigation:

Cost overruns and delays, a long-standing problem in construction. Flyvbjerg et al. (2003) showed that cost overruns in infrastructure projects are common and large, and Love et al. (2012) argue that early estimates are often too optimistic. The Philippine studies report the same pattern. Layno and Famadico (2024) describe the factors behind cost and time overruns in public infrastructure, Sumail et al. (2025) link underestimated costs, rising material prices and slow procurement to overruns, Begay (2024) ranks material delivery and agency decision delays among the main causes, and Bascon et al. (2023) surveyed contractors on 63 factors in vertical construction. Yet these studies only identify what goes wrong, since they rely on surveys of perceived causes and do not predict the cost of a particular project. Arkidom Development Corporation, an architecture, urban planning and construction firm in Quezon City with a small team, needs an early cost estimate for each new project and an early warning when a project is likely to run over budget or schedule.

Machine learning, already used to estimate construction cost early in a project. Hashemi et al. (2020), Elmousalami (2020), Sadikoglu and Demirkesen (2025) and Abed et al. (2022a) review

these methods and note that they need an adequate amount of data. In the Philippines, Roxas et al. (2019) modeled road project cost with a neural network on 41 projects, and Manalindo et al. (2024) predicted the total cost of low- to mid-rise buildings from building elements. Wang et al. (2012) predicted cost and schedule success using neural network ensembles and support vector machines, Gondia et al. (2020) predicted delay risk, and Abed et al. (2022b) compared ridge, LASSO, k-nearest neighbors and random forest on road project records. Random forests (Breiman, 2001) and gradient boosting (Chen & Guestrin, 2016) are widely used for tabular data like project records.

The system, a predictive analytics tool for one firm. Its purpose is to estimate the cost of a new project and to flag the projects at risk of overrunning. The modules of the system are as follows:

Project Data Preparation: imports the firm's project records, cleans them and removes client names and other identifiers.

Cost Estimation: predicts the project cost from the scope, floor area, building type, materials, bill of quantities lines and location, using a linear regression baseline, random forest and gradient boosting.

Overrun Risk: classifies whether a project is likely to exceed its budget or schedule, using the final cost against the contract and the actual dates against the plan.

Estimator Interface: lets the firm enter a new project and see the estimated cost, the range and the main factors.

Evaluation Report: compares the models using MAPE, RMSE and R squared with cross-validation.

To conclude, the study aims to give a small firm an estimate and an early warning built from its own history, which the Philippine studies found so far do not provide.

2.0 Background and Rationale of the Project/System:

Arkidom Development Corporation in Quezon City, a private firm and the client of this study. The system will be built using the firm's own project records. For this term, the adviser requires that the data comes from a private client, starts in 2021 or later and has at least 5,000 rows. A firm of this size may have only a few finished projects, and the published Philippine studies used only 30 to 41 projects, so the group will ask whether the firm keeps bill of quantities line items or cost line items across projects, since the 5,000 rows are realistic only at that level of detail. [To confirm with the client: how many projects since 2021, the record format, who approves sharing data and whether anonymized data is acceptable.] The focus of the study is predictive analytics under the Data Science specialization, and the expected outcome is a working cost estimator and overrun warning tool for the firm, together with a documented comparison of methods on a small private dataset.

The reason for developing the system is simple. The Philippine cost estimation studies found use

public road or building projects, and none looks at a small private architecture and construction firm's own records, while the Philippine overrun studies are surveys rather than predictive models. An estimator built on the firm's own history can support quotations and give early warnings. On ethics, the study follows the Data Privacy Act of 2012 (Republic Act No. 10173): client names and identifiers are removed before the data reaches the group, no personal data is kept in the repository, and no human participants or web scraping are involved. The records are obtained with the firm's written permission, and the results are presented without naming clients.

3.0 Statement of Objectives:

General objective: To develop and evaluate a project cost estimation and overrun prediction system for Arkidom Development Corporation in Quezon City using gradient boosting and baseline regression methods.

Specific objectives:

1. To obtain, clean and anonymize the firm's project and cost records from 2021 onward (at least 5,000 rows of cost line items or equivalent) by the end of Thesis 2, when the project is about 70 percent complete.
2. To build a linear regression baseline, random forest and gradient boosting models for project cost, and to compare them using cross-validation with MAPE, RMSE and R squared, with the target that the best model's MAPE is lower than the baseline's by at least 10 percent (the target will be revised after the data check).
3. To develop an overrun risk classifier for budget and schedule overruns and to report its precision and recall on completed projects.
4. To deliver an estimator interface and to test it with the firm's staff by Thesis 3, when the system and the analysis are finished.

4.0 Scope and Limitations of the Study:

The study covers one firm, Arkidom Development Corporation in Quezon City, and uses its project records from 2021 onward. It estimates the project cost and predicts the risk of budget and schedule overruns for the architectural and construction projects of the types in the records. The inputs are project characteristics that are available at the start of a project, such as the scope, floor area, building type, materials and location.

However, the study does not cover design quality, structural analysis, contractor selection, procurement, real-time site monitoring or other firms. The results apply to this firm's type of projects and may not hold for others. With a small number of projects the estimates will have wide ranges,

and the accuracy depends on how complete the records are. Client details are anonymized.

5.0 Target Beneficiaries:

The groups that will benefit from the study are as follows:

Arkidom owners and project managers: get faster and better supported cost quotations, and an early warning when a project is likely to overrun.

Architects, engineers and estimators of the firm: have a consistent estimate to check their own figures against.

Clients of the firm: get more realistic budgets and fewer surprises, which is an indirect benefit.

Other small construction and design firms: may adopt a documented method.

Future students and researchers: have a local case study on machine learning cost estimation with a small private dataset.

6.0 SDG Category Number:



SDG Number: 9 and 11

SDG Title: Industry, Innovation and Infrastructure; Sustainable Cities and Communities

Justification:

SDG 9, Industry, Innovation and Infrastructure (target 9.1, quality, reliable and resilient infrastructure). Better cost estimates help projects finish within budget. The study also supports SDG 11, Sustainable Cities and Communities, because the firm builds in Quezon City. [The group should confirm the SDG with the adviser.]

7.0 References:

Abed, Y. G., Hasan, T. M., & Zehawi, R. N. (2022a). Machine learning algorithms for constructions cost prediction: A systematic review. International Journal of Nonlinear Analysis and Applications, 13(2), 2205-2218. Retrieved from https://ijnaa.semnan.ac.ir/article_6693.html

Abed, Y. G., Hasan, T. M., & Zehawi, R. N. (2022b). Cost prediction for roads construction using machine learning models. International Journal of Electrical and Computer Engineering Systems, 13(10), 927-936. doi:10.32985/ijeces.13.10.8

Bascon, R. D. D., Gangcuangco, R. L., Carreon, A. V., Morales, M. D., Alejandrino, A. J. R., Nuqui, N. I., ... Tongol, J. V. G. (2023). Investigation on the factors influencing time and cost overrun in vertical construction in Pampanga. IRE Journals, 6(12), 730-752. Retrieved from https://www.irejournals.com/paper-details/1704711

Begay, B. C. (2024). Analysis of construction delays in public sector projects at Apayao State College in the Philippines. Library Progress International, 44(3). Retrieved from https://bpasjournals.com/library-science/index.php/journal/article/view/2868

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32. doi:10.1023/A:1010933404324

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794). New York, NY: ACM. doi:10.1145/2939672.2939785

Elmousalami, H. H. (2020). Artificial intelligence and parametric construction cost estimate modeling: State-of-the-art review. Journal of Construction Engineering and Management, 146(1). doi:10.1061/(ASCE)CO.1943-7862.0001678

Flyvbjerg, B., Skamris Holm, M. K., & Buhl, S. L. (2003). How common and how large are cost overruns in transport infrastructure projects? Transport Reviews, 23(1), 71-88. doi:10.1080/01441640309904

Gondia, A., Siam, A., El-Dakhakhni, W., & Nassar, A. H. (2020). Machine learning algorithms for construction projects delay risk prediction. Journal of Construction Engineering and Management, 146(1). doi:10.1061/(ASCE)CO.1943-7862.0001736

Hashemi, S. T., Ebadati, O. M., & Kaur, H. (2020). Cost estimation and prediction in construction projects: A systematic review on machine learning techniques. SN Applied Sciences, 2(10), 1703. doi:10.1007/s42452-020-03497-1

Layno, J. D. J., & Famadico, J. J. F. (2024). Cost and time overrun of public infrastructure project in the Philippines: Inhibiting factors and mitigating measures. International Journal of Multidisciplinary: Applied Business and Education Research, 5(12), 5360-5370. doi:10.11594/ijmaber.05.12.30

Love, P. E. D., Edwards, D. J., & Irani, Z. (2012). Moving beyond optimism bias and strategic misrepresentation: An explanation for social infrastructure project cost overruns. IEEE Transactions on Engineering Management, 59(4), 560-571. doi:10.1109/TEM.2011.2163628

Manalindo, A. Y. L., Silva, D. L., Diona, R. L., & de Jesus, K. L. M. (2024). Artificial neural network prediction of total construction cost using building elements for low- to mid-rise buildings. In Lecture Notes in Civil Engineering (pp. 441-452). Springer. doi:10.1007/978-981-97-5910-1_34

Republic Act No. 10173. (2012). Data Privacy Act of 2012. Retrieved from https://www.lawphil.net/statutes/repacts/ra2012/ra_10173_2012.html

Roxas, C. L. C., Roxas, N. R., Cristobal, J. B., Hao, S. E. M., Rabino, R. M. G., & Revalde, F. M. (2019). Modeling road construction project cost in the Philippines using the artificial neural network approach. In 2019 IEEE 11th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM). doi:10.1109/HNICEM48295.2019.9072820

Sadikoglu, E., & Demirkesen, S. (2025). Review of machine learning and artificial intelligence use for cost estimation in construction projects. Computing in Construction, 6. doi:10.35490/ec3.2025.363

Sumail, N. A. G., Taclendo, C., Asturias, R., & Pedroso, J. F. (2025). Delays and cost overruns in infrastructure projects in Cotabato City, BARMM: Causes, impacts, and mitigation strategies. International Journal for Multidisciplinary Research, 7(3). Retrieved from https://www.ijfmr.com/research-paper.php?id=46966

United Nations. (2015). Transforming our world: The 2030 agenda for sustainable development. Retrieved from https://sdgs.un.org/2030agenda

Wang, Y.-R., Yu, C.-Y., & Chan, H.-H. (2012). Predicting construction cost and schedule success using artificial neural networks ensemble and support vector machines classification models. International Journal of Project Management, 30(4), 470-478. doi:10.1016/j.ijproman.2011.09.002

Approved by:

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Panel Member 2 / Date

Noted by:

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